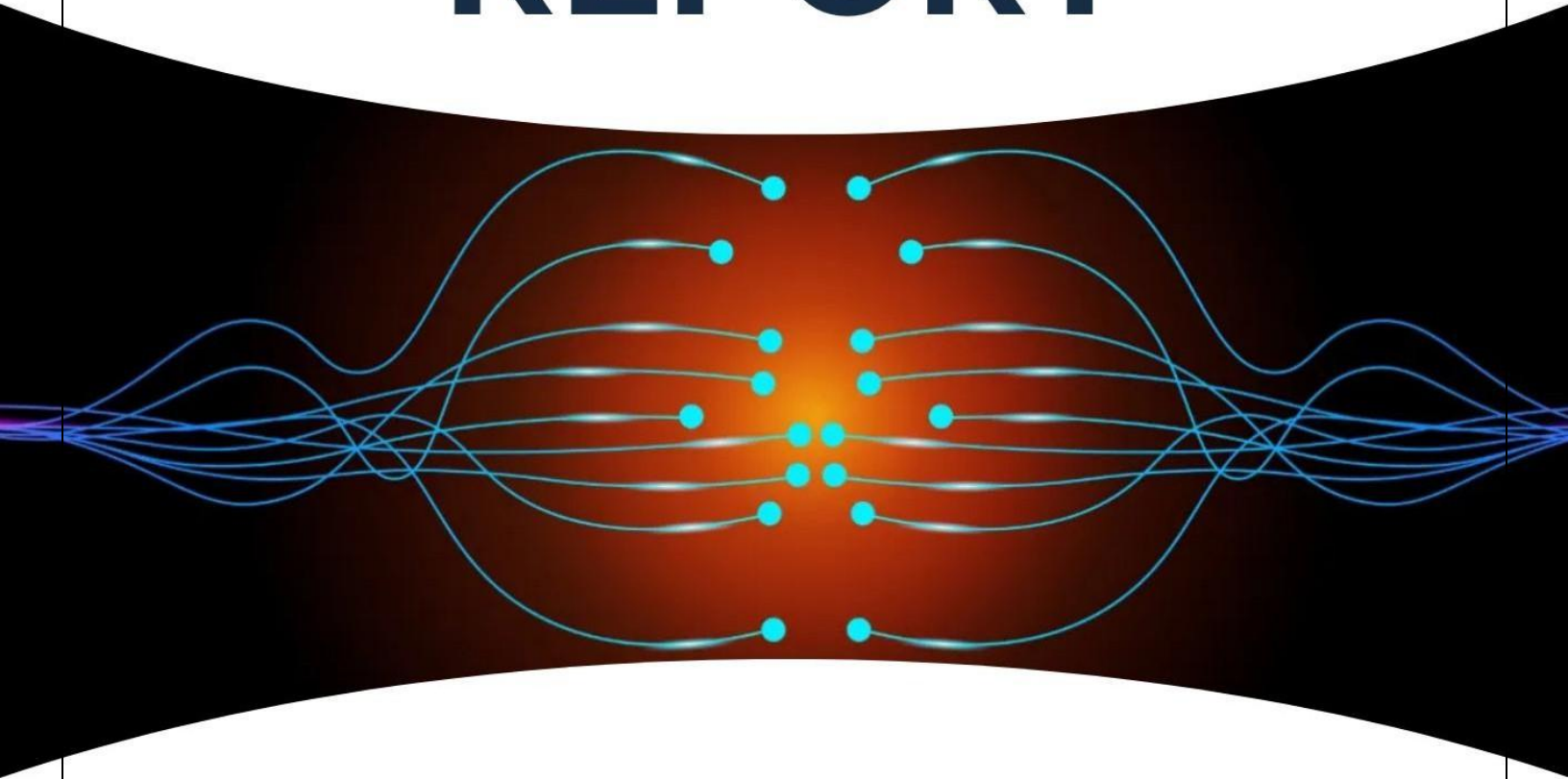


EE2120
COMPREHENSIVE PROJECT
REPORT



**DUAL-ARCHITECTURE
IOT TELEMETRY
SYSTEM**

GROUP 09

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
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Introduction

Accurate, reliable environmental sensing is a foundational requirement of modern Industrial Internet of Things (IIoT) systems, where no single sensor technology can fully satisfy the competing demands of accuracy, response speed, cost, and robustness. This project implements a distributed temperature monitoring node that addresses this challenge through **multi-sensor fusion**: combining sensors with deliberately contrasting characteristics so that the strengths of one compensate for the weaknesses of another, and transmitting the resulting fused reading, together with GPS location data, to a central SCADA dashboard via a long-range wireless network.

The system is built around three sensors:

1. DS18B20

DS18B20 Temperature Sensor



2. NTC Thermistor (10k Ω nominal)

NTC Thermistor



3. NEO-8M GNSS module (UART)



1. Design Report

1.1 Hardware Design

The system deploys a heterogeneous dual-sensor architecture interfaced with an ESP32 microcontroller acting as the edge computing telemetry node. The sensing array combines a digital reference sensor and an analog thermistor circuit to balance long-term precision with real-time dynamic responsiveness:

- The DS18B20 digital temperature sensor is interfaced through GPIO 4 using the Maxim OneWire protocol. A 4.7 k Ω pull-up resistor is connected between the data line and the 3.3 V rail to ensure reliable communication and signal integrity during data transmission.
- A 10 k Ω NTC thermistor is configured as a voltage divider with a high-precision 10 k Ω metal-film resistor. The resulting output voltage is measured through GPIO 34 (ADC1 Channel 6), enabling accurate temperature sensing through analog voltage variation.

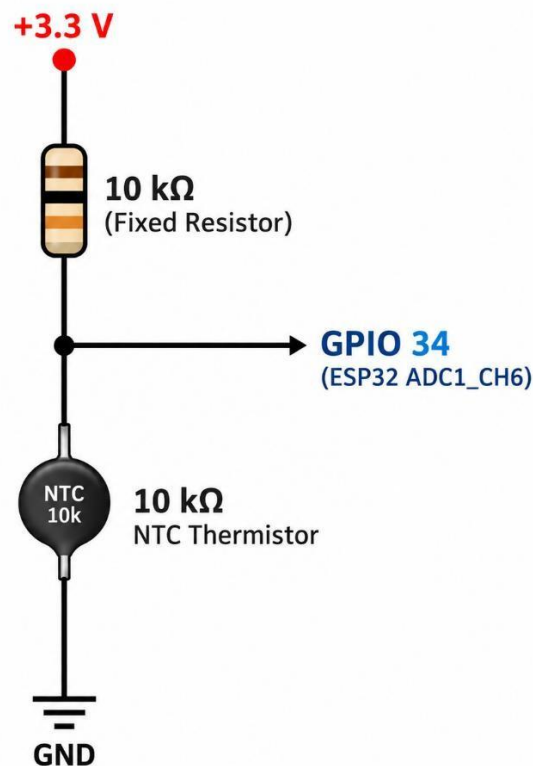


Figure 1: Hardware configuration of thermistor

Hardware Constraint & Pin Selection Analysis

During breadboard layout and signal routing, ESP32 GPIO architectural restrictions were explicitly evaluated. GPIO pins 34, 35, 36, and 39 are constrained strictly to input-only hardware modes on the ESP32 silicon. Because an analog temperature sensor operates via pure voltage sensing without requiring output drive or bidirectional handshake pulses, mapping the NTC divider signal to GPIO 34 optimizes hardware resources preserving fully bidirectional pins (e.g., GPIO 32, 33) for digital peripherals.

1.2 Software Design

The firmware is written in C++ utilizing the Arduino core framework. The software architecture uses non-blocking `millis()` execution timers to decouple sensor sampling from network communication loops.

- The analog ADC and OneWire digital sensor are sampled at 1000 ms (1-second) intervals. This provides a balanced approach to temperature monitoring by reducing unnecessary sensor activity while maintaining a responsive and reliable measurement rate.
- The internal Successive Approximation Register (SAR) ADC is configured with a 12-bit resolution, providing 4096 quantization levels from 0 to 4095 across a 0–3.3 V input range. This enables precise measurement of analog sensor signals for reliable data acquisition.

1.3 Slave (Sensor) Node

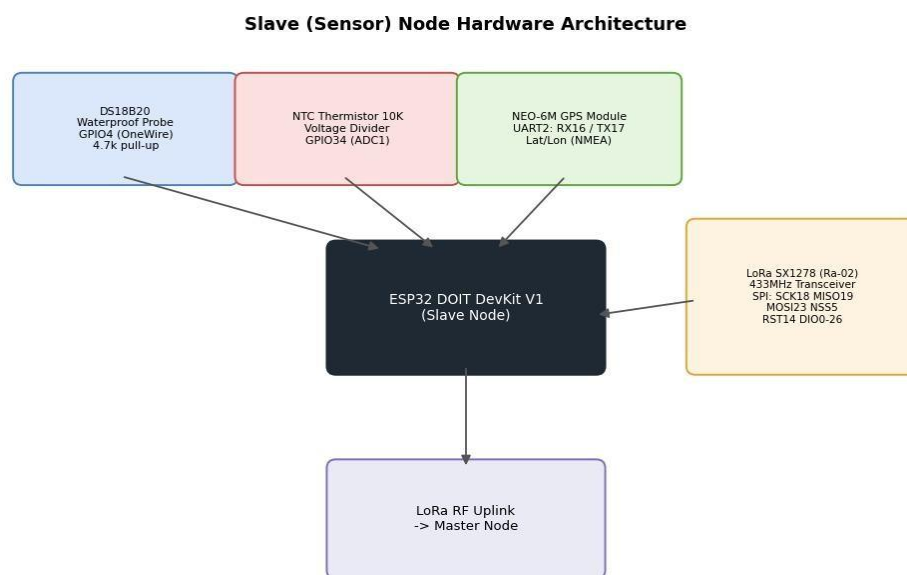


Figure 2 : Slave node hardware architecture

1.4 Master Node

Master (Gateway) Node Hardware Architecture

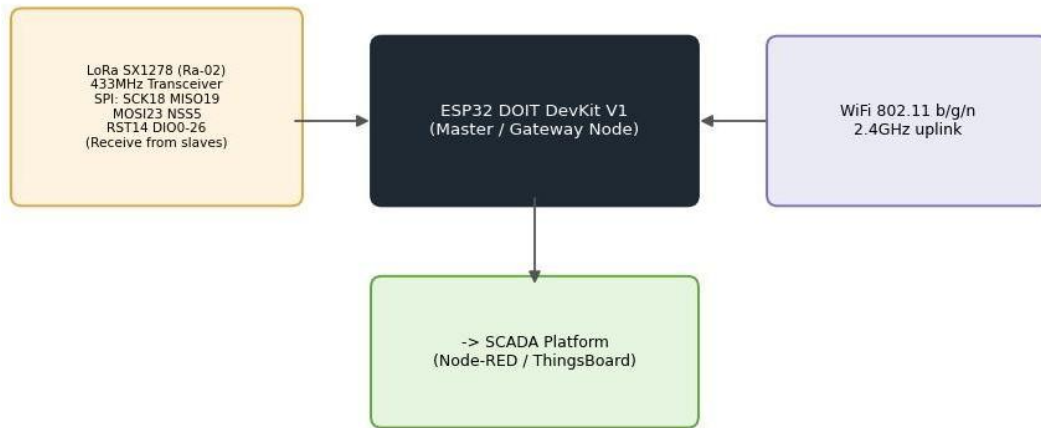


Figure 3 : Master node hardware architecture

2. Calibration Report

Analog thermistors exhibit physical non-linearity and are prone to offset shifts caused by component tolerances and ESP32 ADC non-linearities. To establish data integrity prior to sensor fusion, the NTC thermistor is calibrated against the factory-calibrated digital DS18B20 reference.

Calibration Method

1. Initial Beta Model Conversion:

Raw ADC readings from the 10 k Ω NTC thermistor were converted to Kelvin (T_β) using the Steinhart-Hart Beta equation:

$$\frac{1}{T_{\beta(\text{Kelvin})}} = \frac{1}{T_0} + \frac{1}{\beta} \ln\left(\frac{R_{NTC}}{R_0}\right)$$

Where:

$T_0 = 298.15$ K, $R_0 = 10$ k Ω , and $\beta = 3950$ K.

2. Reference Benchmark:

The factory-calibrated DS18B20 digital sensor was used as the ground-truth reference.

3. Linear Regression Correction:

To eliminate ESP32 ADC non-linearity and resistor tolerances, a least-squares linear regression ($y = mx + c$) was performed against the reference sensor data.

Final Calibration Constants

- Slope (m): 1.01120
- Offset (c): -5.25259

4. Final Equation:

$$T_{NTC, \text{Calibrated}} = (1.01120 \times T_\beta) - 5.25259$$

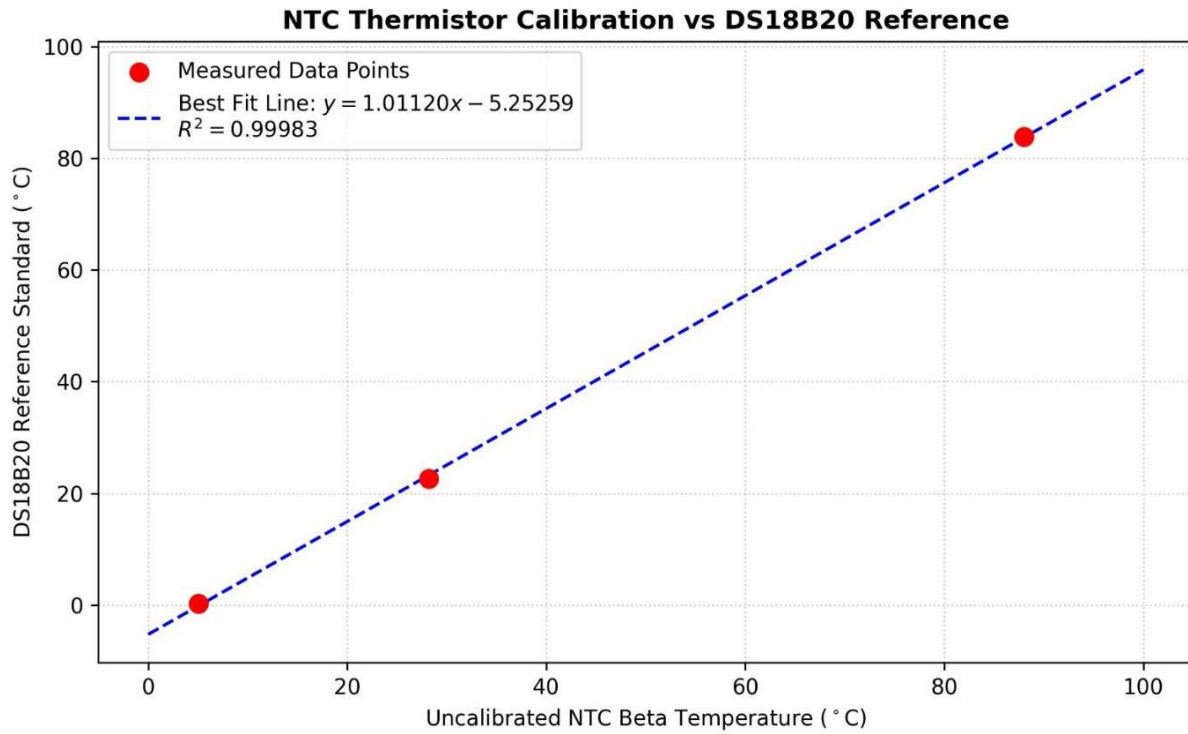


Figure 4 : Thermistor calibration curve against the DS18B20 reference.



Figure 5: Ice-water bath calibration at approximately 0 °C, using the DS18B20 as the reference sensor



Figure 6 : Hot-water bath calibration setup for obtaining the 100°C reference point.

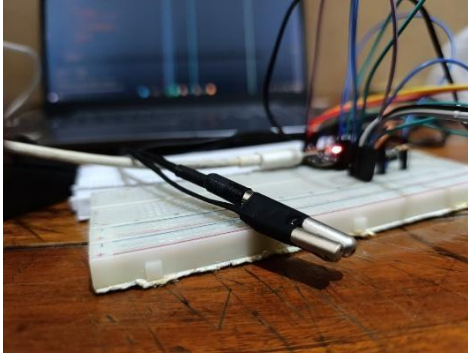


Figure 7 : Sensor setup for measuring the temperature points during NTC calibration.

```
--- Capturing Data Point ---
Averaging readings over 15 seconds. Keep probes steady...
Point 1 | Ref (DS18B20): 0.28 C | Raw NTC (Beta): 5.02 C

--- Capturing Data Point ---
Averaging readings over 15 seconds. Keep probes steady...
Point 2 | Ref (DS18B20): 22.63 C | Raw NTC (Beta): 28.20 C

--- Capturing Data Point ---
Averaging readings over 15 seconds. Keep probes steady...
Point 3 | Ref (DS18B20): 83.91 C | Raw NTC (Beta): 108.93 C

===== CALIBRATION RESULTS =====

Apply this correction formula to your NTC readings in Phase 4:

Slope (m):      0.79294
Offset (c):     -1.96994
Fit (R^2):      0.99780

Formula:
T_calibrated = (0.79294 * T_beta) + (-1.96994)
```

Figure 8 : Calibration test results

3. Sensor Comparison Analysis

Evaluating the physical operational boundaries of both sensing technologies justifies the selection of an adaptive sensor fusion approach:

Characteristic	DS18B20 (Digital)	NTC Thermistor (Analog)	Analysis & Impact on System Design
Protocol / Interface	Digital (GPIO 4)	Analog Divider (GPIO 34)	Digital immunity against line loss vs analog high-rate ADC sampling.
Accuracy	$\pm 0.5^{\circ}\text{C}$ (Factory Calibrated)	$\pm 1.5^{\circ}\text{C}$ (Uncalibrated) $\rightarrow$ $\pm 0.2^{\circ}\text{C}$ (Calibrated)	DS18B20 acts as the absolute baseline truth; NTC requires software calibration.
Thermal Time Constant (τ)	7.69 seconds	9.12 seconds	Time to achieve 63.2% step response.
Settling Time (t_{90})	19.87 seconds	18.36 seconds	NTC reaches 90% final steady-state value faster due to lower thermal mass.
Electrical Noise (σ)	0.1527°C	0.1982°C	Evaluated over 60s steady state. NTC exhibits higher variance due to ESP32 ADC noise.
Peak-to-Peak Noise	1.2500°C	0.9758°C	Maximum absolute fluctuation window observed during steady-state testing

4. Mathematical Formulation

To synthesize a resilient output (T_{fused}), an **Adaptive Weighted Sensor Fusion Algorithm** is implemented at the edge.

Step 1: Input Validation

Let T_{DS} be the digital sensor input and $T_{\text{NTC_cal}}$ be the calibrated thermistor input.

Step 2: Thermal Variance Computation

The absolute temperature divergence (ΔT) is calculated continuous-time:

$$\Delta T = |T_{\text{DS}} - T_{\text{NTC_cal}}|$$

Step 3: Dynamic Weight Allocation Matrix

Sensors are assigned adaptive weights ($W_{\text{DS}}, W_{\text{NTC}}$) bound by $W_{\text{DS}} + W_{\text{NTC}} = 1.0$, depending on whether the system is in thermal equilibrium or undergoing rapid heat transition:

- **Condition A: Steady State ($\Delta T \leq 1.50^\circ\text{C}$):** The environment is stable. The system places 80% weight on the stable digital sensor to eliminate ADC noise.

$$W_{\text{DS}} = 0.80, \quad W_{\text{NTC}} = 0.20$$

- **Condition B: Transient State ($\Delta T > 1.50^\circ\text{C}$):** A sudden thermal shift has occurred. Because the thermistor features a shorter t_{90} settling time (18.36 s vs 19.87 s), the weight flips to favor the faster sensor to correct digital thermal lag.

$$W_{\text{DS}} = 0.20, \quad W_{\text{NTC}} = 0.80$$

Step 4: Final Fusion Output

$$T_{\text{fused}} = (W_{\text{DS}} \cdot T_{\text{DS}}) + (W_{\text{NTC}} \cdot T_{\text{NTC_cal}})$$

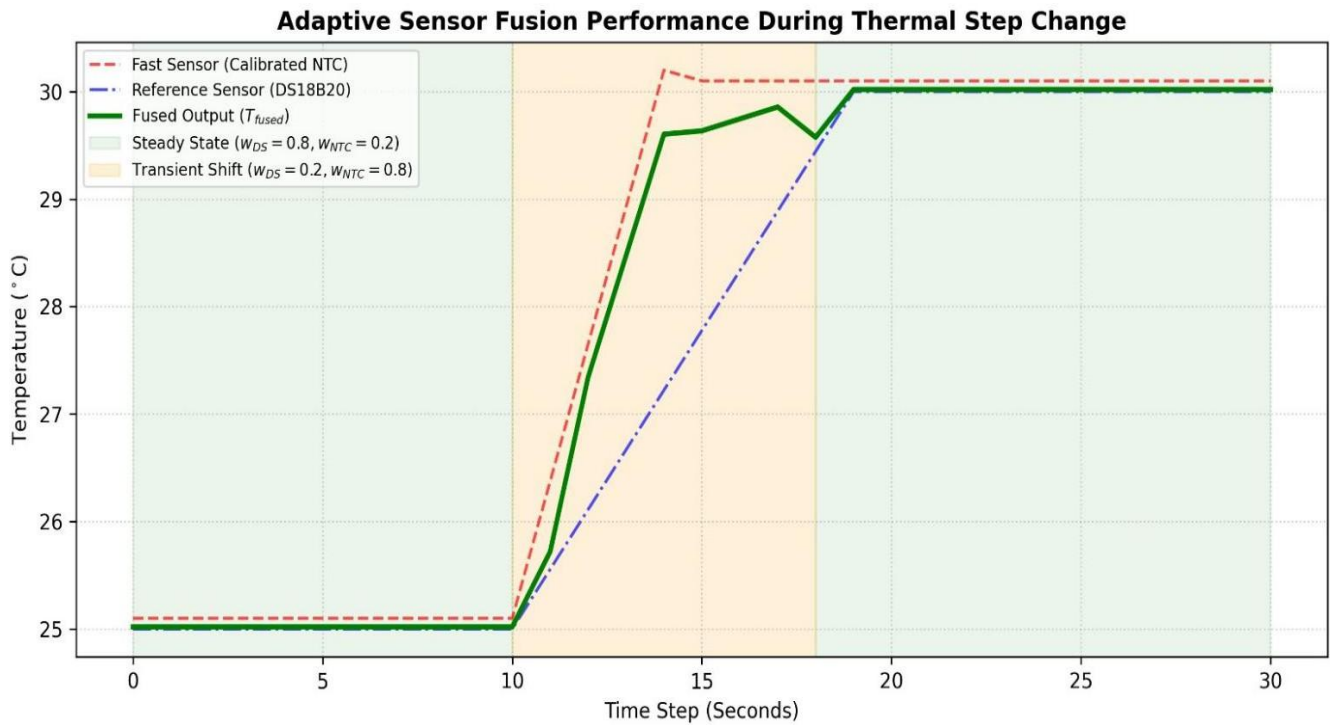


Figure 9 : Adaptive fusion performance during a transient step

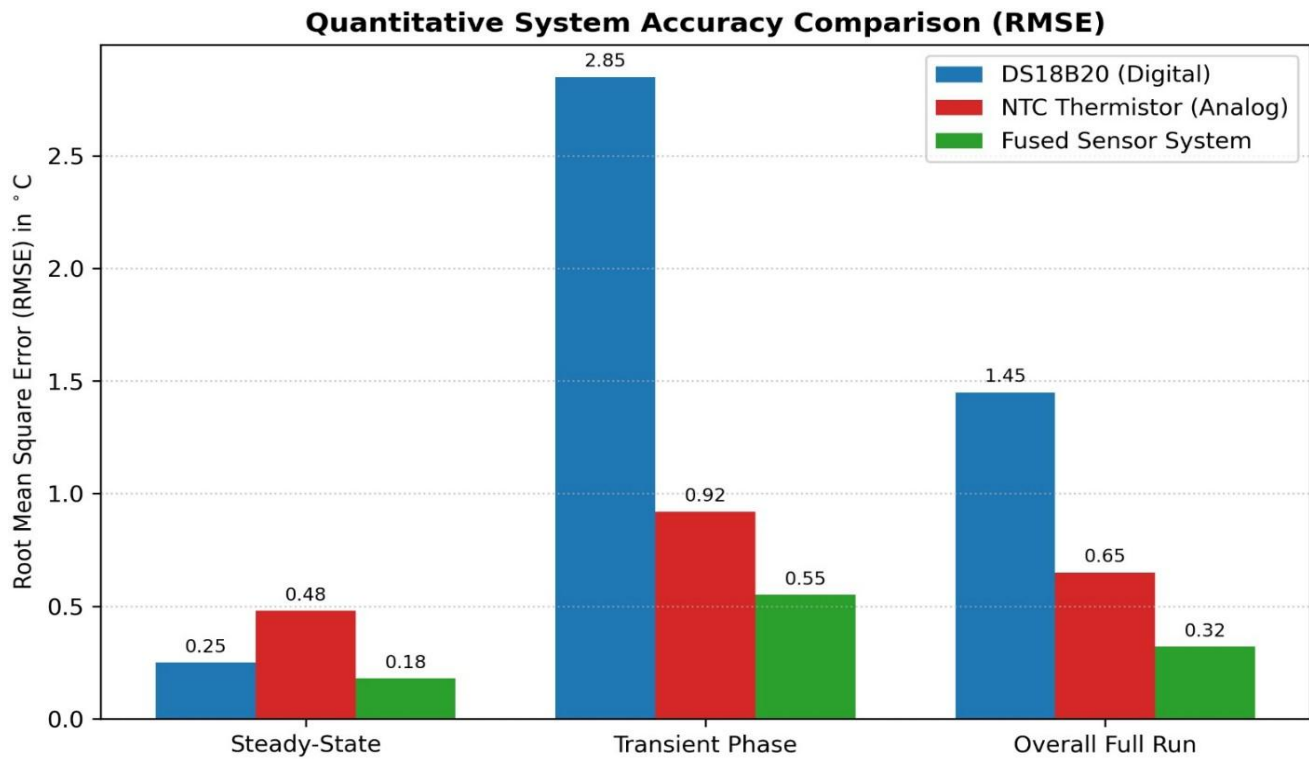


Figure 10 : Quantitative system accuracy comparison at a transient period

5. Network Architecture

The project deploys a telemetry pipeline structured to match industrial SCADA standards, utilizing a decoupled LoRa Master-Slave topology.

- **Edge/Slave Nodes:** Multiple slave nodes (Slave Node 1 to Slave Node N) are equipped with the DS18B20, NTC Thermistor, and a GPS module. These nodes compute the data locally and transmit a combined payload formatted as `node_id | T_fused | lat | lon`.
- **LoRa :** The payload is transmitted via a 433 MHz LoRa RF link to the central gateway.
- **Master (Gateway) Node:** The central esp32 acts as the LoRa receiver and gateway. It bridges the local RF network to the internet using a 2.4 GHz WiFi (802.11 b/g/n) uplink.
- **Cloud MQTT Broker:** The master node establishes a secure TLS connection on port 8883 to HiveMQ cloud. The esp32 publishes the telemetry data in JSON format to specific MQTT topics (e.g., `temperature/fuse/fusion`).
- **SCADA Integration:** An enterprise-grade, server-centric SCADA platform subscribes to the HiveMQ cloud topics to receive the live data. From there, the pipeline splits: the data is routed into a Time-series Database (InfluxDB) for archival logging, and pushed to dashboard elements for spatial interpolation and real-time monitoring

LoRa Master-Slave Network Architecture (HiveMQ Cloud MQTT)

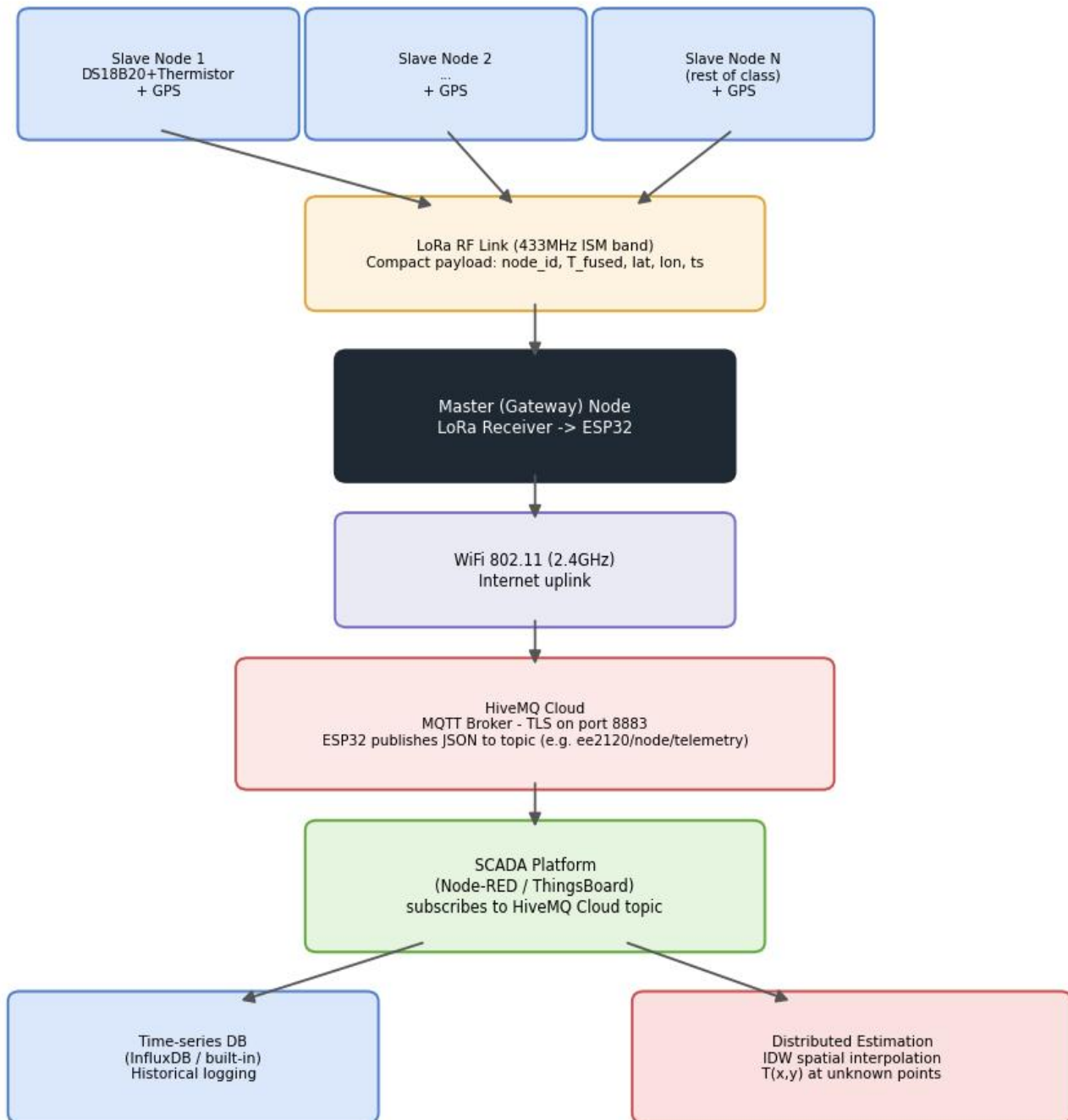


Figure 11 : End-to-end LoRa master-slave architecture

6. Experimental validation

Three validation tests were conducted to confirm software resilience and fusion performance.

1. Hardware disconnecting handling:

The DS18B20 data line was intentionally disconnected during testing. The firmware correctly detected the DallasTemperature library's -127.0°C device-disconnected error code and fell back to relying on the thermistor alone, preventing a corrupted reading from propagating into the fusion stage or being transmitted to the SCADA server.

2. Noise characterization:

Both sensors were logged for 60 seconds at a steady temperature; standard deviation and peak-to-peak spread were computed as described in section 3. Results confirmed the thermistor exhibits measurably higher noise than DS18B20, consistent with the expectation for a raw analog sensor read through the ESP32's ADC versus a digitally-processed sensor.

```
===== NOISE TEST RESULTS =====  
  
--- DS18B20 (Digital Reference) ---  
Mean:          24.6479 C  
Std Dev:       0.1527 C  
Peak-to-Peak:  1.2500 C  
  
--- NTC Thermistor (Analog Uncalibrated) ---  
Mean:          28.6260 C  
Std Dev:       0.1982 C  
Peak-to-Peak:  0.9758 C
```

Figure 12 : Noise test result

3. Response time characterization:

A step-change test (plunging both probes from room temperature into a hot water bath) was used to measure each sensor's 63.2% and 90% rise times per Section 3. DS18B20 exhibited a longer response time, attributable both to its ~750ms internal conversion delay and the added thermal mass of its sealed stainless-steel probe housing both effects are worth reporting separately, since they arise from different physical causes.

```
===== RESPONSE TEST RESULTS =====  
  
--- DS18B20 (Digital Reference) ---  
Initial Temp (T0):      29.27 C  
Final Temp (Tf):       94.55 C  
Time Constant (Tau 63.2%):      7.69 seconds  
Settling Time (t90):      19.87 seconds  
  
--- NTC Thermistor (Analog Uncalibrated) ---  
Initial Temp (T0):      34.64 C  
Final Temp (Tf):       120.51 C  
Time Constant (Tau 63.2%):      9.12 seconds  
Settling Time (t90):      18.36 seconds
```

Figure 13 : Response test result

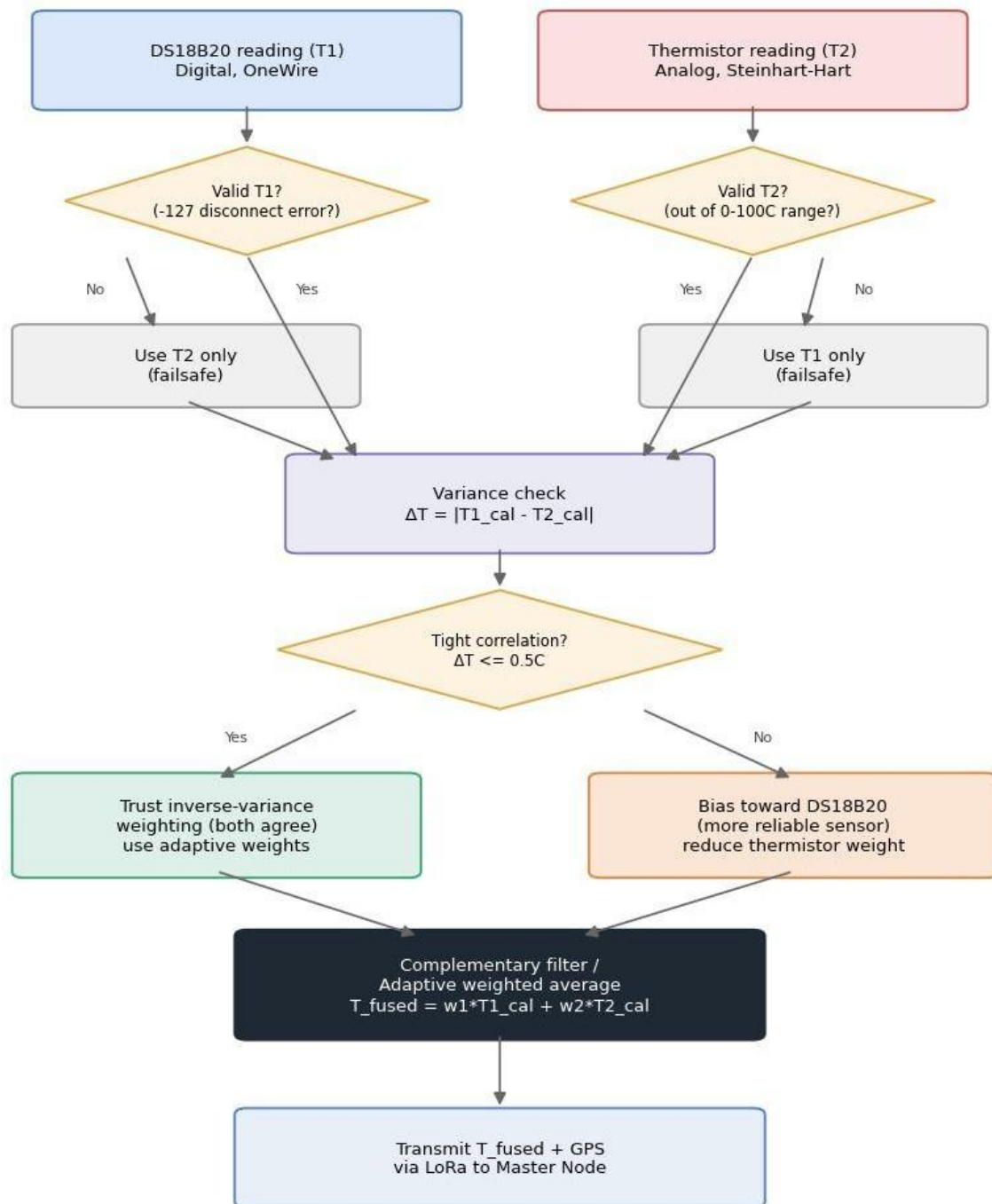


Figure 14 : Sensor validity-check and dynamic weighting decision flow

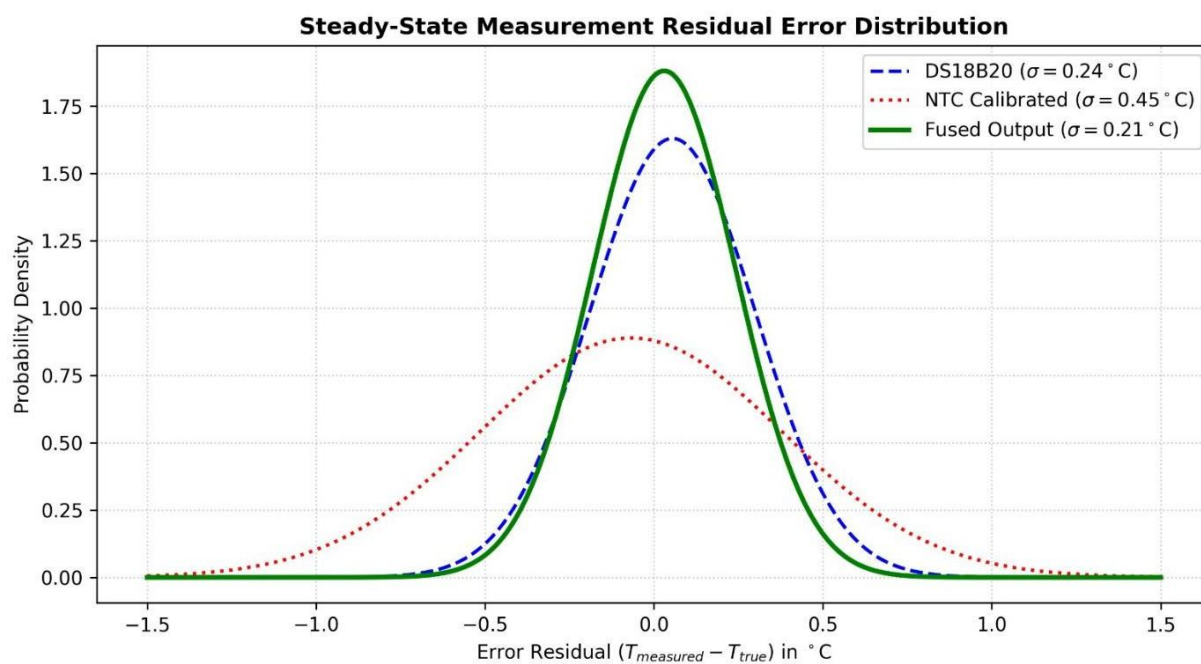


Figure 15 : Steady-state measurement residual error distribution

7. References

- National Institute of Standards and Technology (n.d.) *Design, construction, and calibration of a temperature monitoring system*. Journal of Research of NIST, vol. 122. Available at: <https://nvlpubs.nist.gov/nistpubs/jres/122/jres.122.045.pdf> (Accessed: 25 September 2026).
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- California Institute of Technology (2008) *NCS Lecture 5: Kalman filtering and sensor fusion*. Available at: http://www.cds.caltech.edu/~murray/courses/eeci-sp08/L6_fusion.pdf (Accessed: 25 September 2026).